**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Sequence, Input and Output

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Name one operating-system function.

2. What does software copyright restrict?

A) Using any keyboard
B) Learning to code
C) Copying without permission
D) Saving your own work

3. What is a typical feature of proprietary software?

4. A school needs a large, cheap archive drive. A pupil's only coursework copy is on an SSD. Correct both claims. Explain why. (a) An HDD is always faster and tougher than an SSD. (b) An SSD never fails, so needs no backup.

## Today's objectives

Write programs containing instructions executed in sequence. Use sequence and input in a practical scenario.

## Key vocabulary - Read, cover, recall

**Sequence**

Instructions execute in order, so each value is ready before it is used.

**Input**

The input() command obtains data from a user and can store it in a variable.

**Processing**

Expressions calculate a new value from existing values.

**Output**

print() displays the result of a completed calculation.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **What is Sequence?** | Sequence - Instructions execute in order, so each value is ready before it is used. |
| **In this example** | A bus ticket price app uses input; The input() command obtains data from a user and can store it in a variable. |
| **Follow the effect** | Expressions calculate a new value from existing values. Explain how this changes the eventual program result. |
| **Check understanding** | Output is used for a specific reason: print() displays the result of a completed calculation. |

## I Do - One worked example

**A bus ticket price app is being written. Explain how sequence and input are used to make the program work. [4 marks]**

1. We are designing a bus ticket price app. Identify which data and decisions are essential.

2. Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1].

3. Input [1] - The input() command obtains data from a user and can store it in a variable. [1].

4. Two linked points for each technique give four mark points.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a school uniform total, explain how sequence helps the program. [2 marks]

2. For a two sports scores, explain why input matters. [2 marks]

3. When designing a cinema booking, explain how processing and output should be used. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Write one line to display the name the user enters. Assume the input is "Ava".**

```
name = input("Your name?")
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A shop receipt needs to work correctly. Explain how input and processing would help. [4 marks]

Q4. A pupil is programming a online café bill. Explain the roles of sequence, processing and output in their solution. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Instructions execute in order, so each value is ready before it is used. ☐ T ☐ F
2. Sequence - Instructions execute in order, so each value is ready before it is used. ☐ T ☐ F
3. A bus ticket price app uses input; The input() command obtains data from a user and can store it in a va... ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Memory management, user management, file management, a user interface or peripheral management. ☐

**Q2** C - Copying without permission ☐

**Q3** Source code is normally controlled by its owner and users receive specified usage rights. ☐

**Q4** (a) An HDD can give lots of space at low cost per GB. Check the device and task. (b) An SSD can fail through faults or wear, so keep backups. No device is best at everything. ☐

## Code First - Model

```
name = input("Your name?")
print(name)
```

**Output: Ava (assuming input Ava)**

## We Do - Check the model

1. Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1]. ☐

2. Input [1] - The input() command obtains data from a user and can store it in a variable. [1]. ☐

3. Processing [1] - Expressions calculate a new value from existing values. [1]. Output [1] - print() displays the result of a completed calculation. [1] ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Sequence [1]; Processing [1]. ☐

**Q3** Input [1] - The input() command obtains data from a user and can store it in a variable. [1]. Processing [1] - Expressions calculate a new value from existing values. [1]. ☐

**Q4** Sequence [1] - Instructions execute in order, so each value is ready before it is used. [1]. Processing [1] - Expressions calculate a new value from existing values. [1]. Output [1] - print() displays the result of a completed calculation. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**